JDScience BTEC Level 3 Chemical Calculations Worksheet

Balanced equations, relative atomic mass and relative formula mass.

Learning aims

- Balance chemical equations by conserving atoms.
- Calculate relative atomic mass from isotopic abundance.
- Calculate relative formula mass (Mr) from atomic masses.
- Show clear working for BTEC-style calculation questions.

Key notes

Balanced equations have the same number of each atom on both sides. Only coefficients in front of formulae may be changed; formula subscripts must not be changed.

Relative atomic mass can be a decimal because it is a weighted mean of all isotopes. Chlorine is about 75% chlorine-35 and 25% chlorine-37, giving Ar = 35.5.

Relative formula mass is calculated by adding the Ar values for all atoms in the formula. Brackets must be multiplied carefully.

Skill	Method	Common error
Balancing equations	Count atoms, add coefficients, check again	Changing formulae instead of coefficients
Relative atomic mass	mass x abundance, add, divide by total abundance	Using a simple average
Mr	Multiply each Ar by number of atoms, then add	Forgetting brackets such as Ca(OH)2

Student tasks

A. Balancing equations

Balance these equations. Use whole-number coefficients where possible.

#	Equation	Balanced answer space
1	Mg + HCl -> MgCl2 + H2	
2	Na + O2 -> Na2O	
3	Ca(OH)2 + HNO3 -> Ca(NO3)2 + H2O	
4	Al + O2 -> Al2O3	
5	C2H6 + O2 -> CO2 + H2O	
6	Fe + Cl2 -> FeCl3	

B. Relative atomic mass

Calculate the relative atomic mass of chlorine from 75% Cl-35 and 25% Cl-37. Explain why the answer is not a whole number.

C. Relative formula mass

Calculate the Mr of: CO2, Ca(OH)2, Mg(NO3)2, Al2(SO4)3 and NaAl(OH)4.

D. Exam-style challenge

Ethane burns in oxygen to form carbon dioxide and water. Balance the equation and explain why balancing equations supports the law of conservation of mass.